15	**Ans: A** There is a decrease in volume from P to Q, hence it is a compression. Bearing in mind that the gradient of the line for adiabatic process is steeper than that of isothermal (due to the increase/decrease in temperature – shift of isotherms), the process from P to Q can be seen to be an adiabatic compression (it occurs rapidly such that heat is not exchanged). Option B - Process Q to R is a isothermal expansion. So the temperature of the gas at state Q is equal to that at state R. Option C – there is a net positive work done by gas for the process PQRP (due to the larger area under the process Q to R. Option D – since ΔU is a function of state, there is no change in internal energy as the gas undergoes the process PQRP.